MILITARY SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET MOTOR, CANOPY JETTISON, MARK 109 MOD 0 (LOADED ASSEMBLY)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the Mark 109 Mod 0 Canopy Jettison Rocket Motor described herein shall consist of this document and the latest issue of MIL-A-85097.

1. SCOPE

1.1 Scope. This specification sheet covers requirements unique to the Canopy Jettison Rocket Motor, Mark 109 Mod 0 (loaded assembly).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specification is in addition to those listed in the cover specification, MIL-A-85097, and form a part of this specification sheet to the extent specified herein.

SPECIFICATIONS

Military

MIL-B-5087 Bonding, Electrical and Lighting Protection, For Aerospace Systems

2.1.2 Other Government documents, drawings, and publications. The following drawing forms a part of this specification sheet to the extent specified herein.

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

1507AS100 Rocket Motor, Canopy Jettison, Mk 109 Mod 0

3. REQUIREMENTS

3.1 General. The requirements in this specification sheet unless in direct conflict with MIL-A-85097 shall be considered additional, not substitute. The rocket motor described herein shall conform to Drawing 1507AS100 and the weight shall be not greater than 3.50 pounds.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Preproduction.

3.2.1 Propellant aging. When tested in accordance with MIL-A-85097, the propellant shall meet the limit requirements specified by the contractor and which are acceptable to the Government.

3.3 Preproduction/production.

3.3.1 Propellant. The propellant shall conform to the contractor's in-house propellant specification when tested for the data of MIL-A-85097. The contractor shall make available the results of the tested values and the in-house specification limits for comparison (see 6.2.2).

3.3.2 Ballistic performance requirements. The ballistic performance requirements shall conform to table I at both $-40° \pm 5°F$ and $160° \pm 5°F$.

TABLE I. Ballistic performance requirements.

Characteristic	Minimum	Maximum
Total impulse (lb-s)	184	212
Action time (ms)	-	170
Thrust (lb)	-	4300
Time to 70% maximum rocket motor thrust (ms)	-	20

4. QUALITY ASSURANCE PROVISIONS

4.1 Preproduction/production.

4.1.1 Conditioning temperature. The conditioning temperatures and tolerances shall be as follows:

a. Cold: $-40° \pm 5°F$
b. Hot: $160° \pm 5°F$

4.1.2 Conditioning period. Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: 4 hours, minimum
b. Temperature cycling: 5 hours, minimum

4.1.3 Radiographic inspection. All components or assemblies shall be radiographed in two views 90° apart.

4.1.4 Hydrostatic proof test. Each motor case shall be hydrostatically tested as a pressure vessel to 6366 psi for 15 seconds.

4.1.5 Leakage test. Each rocket motor assembly shall be subjected to a dry leak test to verify compliance with a leakage rate of less than 1×10^{-4} cc/sec air.

4.1.6 Threaded joint torque test. The 1.890-24NS-3 threaded joints (2) of each rocket motor assembly shall withstand 400-inch pounds torque without breakaway, when tested in accordance with MIL-P-83126.

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4.1.7 Electrical bonding resistance test. For each rocket motor assembly selected as a sampling test specimen, the electrical resistance between the shielded mild detonating cord (SMDC) housing and the mounting lugs shall be not greater than 0.0025 ohms, when tested in accordance with MIL-B-5087.

5. PACKAGING

(This section is not applicable to this specification sheet.)

6. NOTES

6.1 Intended use. The rocket motor described herein is intended to be used to eject the canopy during an emergency crew ejection from the F/A/TF-18 aircraft.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Mark and Mod designation of rocket motor

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52-209-3 and -4 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.3.1	Acceptance Test Report	UDI-T-22741A	-

(Data item descriptions related to this specification, and identified in this section will be approved and listed as such in DoD 5000.19L., Vol.II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

Preparing activity:
Navy - AS
(Project 1377-N757)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-85097/13(AS)	AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET MOTOR, CANOPY JETTISON, MARK 109 MOD 0 (LOADED ASSEMBLY)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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